

Lesson 1-1

Development of Information Technology and Social Transformation

Learning Objectives

- Explain the major stages in the development of information technology and their impact on society.
- Give examples of social changes and emerging technologies brought about by information technology, and explain their characteristics.

On an ordinary day, a student in Egypt checks messages on an SNS app, pays for breakfast with a cashless app, joins a lesson through online learning, and orders a book from an e-commerce (EC) shop. Twenty years ago, most of this was not possible. Information technology has developed through a series of stages — from the first computers to cloud computing — and at each stage it has changed the way people communicate, work, learn, and pay.

? Guiding question: How has information technology developed through its major stages, and how has each stage changed society?

✎ Explore (in pairs)

With a partner, look at the five time periods in the table below before you read on. For each period, name one device or service from that era that you still use or hear about today. Then predict which stage changed daily life the most, and give one reason.

Key Fact:
At each stage, information technology introduced a new technology or service and also changed how society communicates, works, and does business.

Key Concepts:

- Moore's Law
- SNS
- e-commerce
- remote work
- online learning
- cashless payment
- edge computing
- autonomous driving
- AR / VR
- quantum computing

📌 Point!

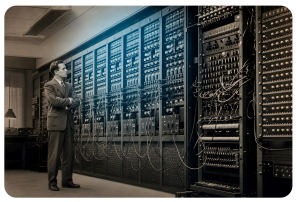
1. The History of Information Technology (IT)

(1) The major stages in the development of information technology are summarized in the table below.

Time period	Major technologies and events	Impact on society
1940s-60s	Birth of the computer (ENIAC, vacuum tubes)	Mainly used for military and scientific computation
1970s-80s	Spread of personal computers (PCs)	Beginning of personal computer use
1990s	Commercialization of the Internet; the Web	Globalization of information; spread of email
2000s	Rise of smartphones (iPhone, etc.)	Explosive spread of mobile Internet
2010s onward	Spread of cloud computing	Large-scale data analysis and AI; "IT as a service" becomes widespread



IT advanced in stages, each smaller and more connected.

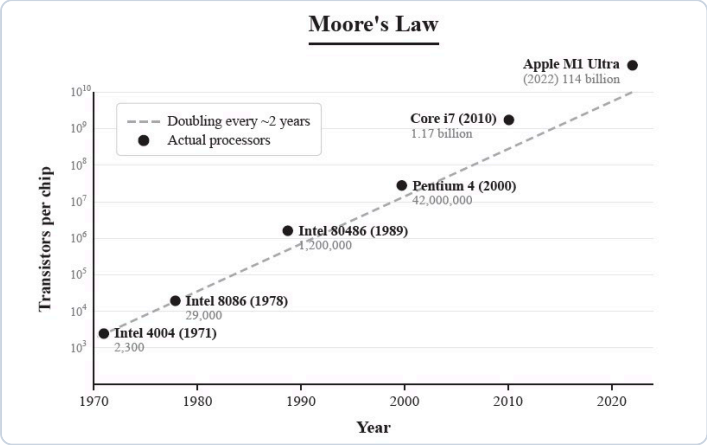


The first computers filled a whole room.

(2) **(Moore's Law)**...the empirical observation that "the number of transistors on an integrated circuit doubles approximately every two years."

This remained largely accurate for many years, and computing power increased dramatically.

However, in recent years, miniaturization of transistors has been approaching a physical limit. If circuits are made any smaller, problems arise such as the quantum tunneling effect, in which electrons slip through barriers, and leakage current, in which current escapes unintentionally, making it difficult to achieve both higher performance and lower power consumption at the same time. In response, new directions for performance improvement are being pursued, such as parallel processing using multiple processor cores, and quantum computers based on the principles of quantum mechanics.



Moore's Law: transistor count over time

2. Social Changes Resulting from Information Technology

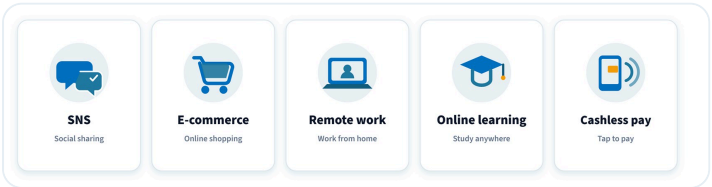
(1) **(SNS (Social Networking Service))**...services that allow users to connect with each other and post and share information. They are highly effective at spreading information rapidly.

(2) **(E-commerce (EC))**...buying and selling goods and services through the Internet.
e.g., online shops such as Amazon and eBay

(3) **(Remote work)**...a working style in which work is performed from home or other remote locations using the Internet.

(4) **(Online learning)**...a learning style in which classes and study materials are delivered using the Internet.

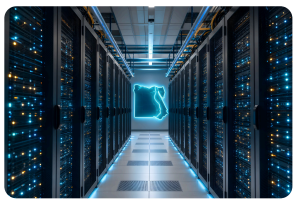
(5) **(Cashless payment)**...a system for making payments using electronic money, QR codes, etc., without using cash.
e.g., credit cards, debit cards, mobile payment apps



Five social changes driven by information technology.

KEY TERMS

Moore's Law — the observation that the number of transistors on a chip roughly doubles about every two years.



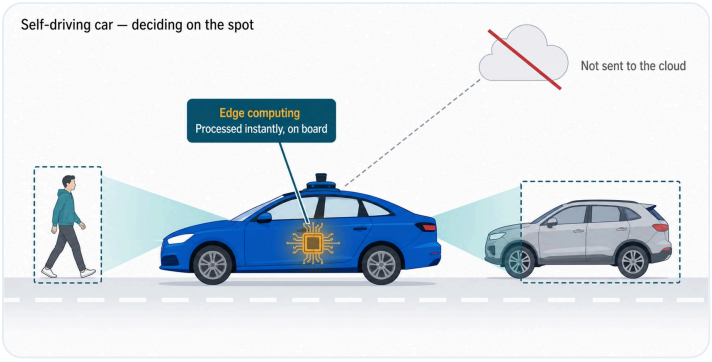
This is "the cloud" — real machines in a real place.

PAUSE & THINK

Of these five changes, which would be hardest to give up — and why?

3. Notable Emerging Technologies

(1) **(Autonomous driving)**...a technology that uses AI to drive a vehicle without human operation. It uses cameras and sensors to recognize the surroundings, makes driving decisions, and controls a vehicle. Because a delay of even 0.1 seconds can lead to an accident, edge computing, in which processing is performed instantly on the vehicle itself rather than sending data to the cloud for judgment, is used.

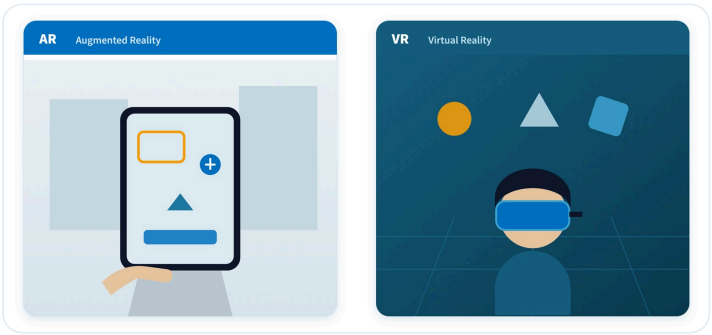


A self-driving car decides on board (edge computing), without waiting for the cloud.

(2) AR / VR

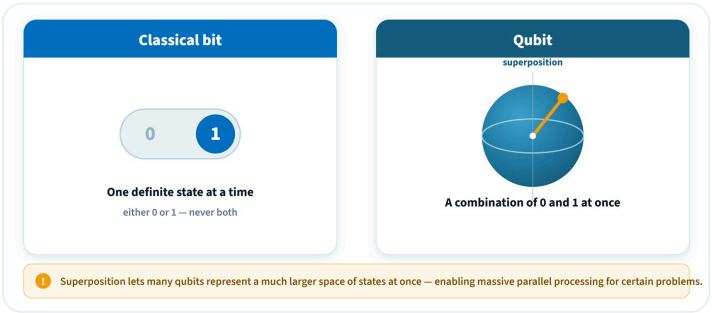
① **(AR (Augmented Reality))**...a technology that overlays digital information on real-world images.

② **(VR (Virtual Reality))**...a technology that allows users to immerse themselves in a virtual space generated by a computer.



AR adds digital layers to the real world; VR replaces it with a virtual one.

(3) **(Quantum computing)**...a technology expected to dramatically speed up computations that are difficult or impossible for traditional computers, by using the principles of quantum mechanics.



A classical bit holds one state; a qubit uses superposition.

KEY TERMS

edge computing — processing data on the device itself, instantly, instead of sending it to the cloud.

PAUSE & THINK

In autonomous driving, why is it necessary to process data instantly on the vehicle side using edge computing, rather than sending the data to the cloud for judgment? Explain your answer.

✎ Worked Example

(1) From the following options (A–D), choose the one that lists the stages of information technology (IT) development in the correct chronological order.

A Birth of the computer → Rise of smartphones → Commercialization of the Internet → Spread of cloud computing

B Birth of the computer → Commercialization of the Internet → Rise of smartphones → Spread of cloud computing

C Commercialization of the Internet → Birth of the computer → Spread of cloud computing → Rise of smartphones

D Rise of smartphones → Commercialization of the Internet → Birth of the computer → Spread of cloud computing

(2) Mark each statement (A–D) with ○ if true or × if false.

A Moore's Law is the empirical observation that "the number of transistors on an integrated circuit doubles approximately every two years."

B Moore's Law has been said to be approaching a physical limit in recent years.

C SNS is highly effective at spreading information rapidly.

D E-commerce (EC) refers to purchasing goods at physical stores using cash.

(3) Match each description (a–c) with the most appropriate technology from the choices below (A–C).

a A technology that overlays digital information on real-world images

b A technology that uses AI to drive a vehicle without human operation

c A technology that allows users to immerse themselves in a virtual space generated by a computer

[Choices] **A** Autonomous driving **B** AR **C** VR

✓ Solution

(1) The correct order is "Birth of the computer (1940s-60s) → Commercialization of the Internet (1990s) → Rise of smartphones (2000s) → Spread of cloud computing (2010s onward)." Therefore, **B**.

(2)

A Correct definition of Moore's Law. Therefore, ○.

B It has been said to be approaching a physical limit in recent years. Therefore, ○.

C SNS is a service highly effective at spreading information rapidly. Therefore, ○.

D E-commerce refers to buying and selling through the Internet, not purchases at physical stores using cash. Therefore, ×.

KEY TERMS

Moore's Law — transistors per chip roughly double every two years.

edge computing — data processed on the device itself, instantly.

(3)

a: Overlaying digital information on real-world images is AR (Augmented Reality). **B**

b: Driving a vehicle without human operation is autonomous driving. **A**

c: Immersing oneself in a virtual space is VR (Virtual Reality). **C**

Try

1 Answer the following questions.

1. What is the name of the empirical observation that "the number of transistors on an integrated circuit doubles approximately every two years"?
2. What is the term for the buying and selling of goods and services using the Internet?
3. What is the term for the working style in which one works from home or other remote locations using the Internet?
4. What is the term for the system for making payments using electronic money, QR codes, etc., without using cash?
5. What is the term for the technology that uses AI to drive a vehicle without human operation?

2 Answer the following questions.

(1) For each of a–d, indicate which of the following is most closely related: **A** Changes in daily life, **B** Changes in industry and the economy, **C** Changes in healthcare and education.

a Taking classes through online learning.

b Paying for purchases using a mobile payment app on a smartphone.

c Sharing photos with friends on SNS.

d A company introducing a work-from-home system.

(2) From the following options (A–D), choose the one that is NOT an appropriate description of an emerging technology.

A Autonomous driving uses AI to drive a vehicle without human operation.

B AR is a technology that overlays digital information on real-world images.

C VR is a technology that dramatically improves the processing speed of a computer.

D Quantum computing is expected to speed up computations that are difficult for traditional computers.

EXAM-STYLE QUESTION

Analyze how the spread of cloud computing (from the 2010s onward) has changed the way information technology is used. **[6] In your answer, refer to:** large-scale data analysis, AI, and "IT as a service".

KEY TERMS

remote work — working from home or another remote location using the Internet.

online learning — classes and materials delivered over the Internet.

⚙ Think as an Engineer — Investigate & Decide

Start from the advantage and concern you named in the Pause & Think note, then investigate further and decide.

1 · Collect data. Survey ten classmates: do they usually pay with cash or with a cashless method (card, mobile app, or QR code)? Record the results in a table and identify the most common response.

2 · Analyze the stakeholders. Using your survey and your own reasoning, note one benefit and one drawback of a fully cashless society for each group below.

Group	Benefit	Drawback
A customer paying		
A small shop owner		
A person with no bank card or smartphone		

3 · Decide. Using your survey and your table, decide: should your community move toward cashless payment? Recommend one step and give two reasons.

Stuck? Start with the customer — one benefit is speed of payment; one drawback is needing a card or a phone.

💡 PAUSE & THINK

Cashless payment is spreading in many countries. If a fully cashless society were realized, choose one advantage and one possible concern, and briefly explain the reason for each.

🌍 In a New Context

A village that has never had Internet access is connected to high-speed Internet and cashless payment for the first time. Using the stages and social changes you studied, predict two ways daily life in the village will change, and identify one new problem the village may face.

KEY TERMS

cloud computing — IT delivered as a service over the Internet.
e-commerce — buying and selling through the Internet.

❓ Lesson Question — Answered

The lesson question asked how information technology developed and how each stage changed society. Information technology advanced through a series of stages rather than in a single step, and each stage changed much more than the technology itself. As computing spread, information became global and then mobile, many everyday activities moved online, and information technology grew into a service that supports large-scale data analysis and AI. Each stage also brought social changes in how people communicate, work, learn, and make payments. Overall, the development of information technology is not only about faster machines; at every stage it has reshaped daily life, industry, and the way society handles information.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

Computers were invented in the 1940s and were initially used mainly for military and scientific computation. Later, in the 1970s and 1980s, (a) became widespread, allowing individuals to use computers. In the 1990s, (b) was commercialized, advancing the globalization of information. In the 2000s, (c) emerged, leading to the explosive spread of mobile Internet.

1. Fill in the blanks (a)–(c).
2. What technology, which spread from the 2010s onward, supports large-scale data analysis and the use of AI?
3. According to Moore's Law, about how often does the number of transistors on an integrated circuit double?

3 Answer the following questions.

(1) From the following options (A–F), choose all that are emerging technologies.

A Email **B** Autonomous driving **C** Personal computers **D** AR **E** VR **F** Quantum computing

(2) From the following options (A–D), choose the one that most appropriately describes the characteristics of SNS.

A A service that allows users to connect with each other and post and share information, and is highly effective at spreading information rapidly.

B A service for buying and selling goods and services through the Internet.

C A working style in which one works from home or other remote locations.

D A system for making payments using electronic money or QR codes without using cash.

REFLECT & CHALLENGE

Reflect: Which stage of information technology do you think will matter most in the next ten years? Give one reason. Was your prediction at the start of the lesson correct? What changed your mind? **Challenge:** Choose one emerging technology from this lesson (autonomous driving, AR, VR, or quantum computing). Suggest one way it could help solve a real problem, and state one risk.

KEY TERMS

SNS — connects users to post and share information.
cashless payment — payment without cash (e-money, QR).

Key takeaway

Remember:
Information technology developed in stages — computers, the Internet, smartphones, and cloud computing. At each stage it introduced a new technology or service and also changed how society communicates, works, learns, and pays.